CS201 Midterm 1 Part B

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Emre SEFER, Olcay Taner YILDIZ

I. Question (LinkedList) (15 points)

Write the method

```
void compressConsecutiveDouble()
```

which compresses consecutive items with the same value into two consecutive values. If a value appears only once consecutively, you only keep the single value without compressing. You are not allowed to use any singly linked list methods. You are allowed to use attributes, constructors, getters and setters. Write the method in the LinkedList class. For instance,

2 2 2 3 3 1 -> 2 2 3 3 1
1 2 3 4 4 2 -> 1 2 3 4 4 2

II. Question (DoublyLinkedList) (15 points)

Write the method

```
boolean IsZigZag()
```

which returns True if doubly linked list is increasing first and then decreasing after some item. This zig zag pattern should appear exactly one time. You may assume all values are distinct. You may assume list has at least 3 values. You are not allowed to use any linked list methods, just attributes, constructors, getters and setters. Write the method in the DoublyLinkedList class.

2 3 4 2 1 -> True
2 4 3 4 5 -> False

III. Question (Stack) (15 points)

Write the method

```
void removeBetweenMinMax()
```

which removes all items between maximum and minimum element, keeping these minimum and maximum elements. Write the function for array implementation. Notice that minimum does not always need to show up before maximum, it can be other way around as well. You may assume there are at least 3 items in the stack. You may assume all elements are distinct. You are not allowed to use any stack methods, just attributes, constructors, getters and setters.

Contents of the stack before running the method:
6 8 9 3 2 1 5 7
Contents of the stack after running the method:
6 8 9 1 5 7

IV. Question (Stack) (15 points)

Write the method

```
void rotateStack(int k)
```

where k is a positive integer representing the number of positions to rotate the stack by, which is implemented using a linked list. After the rotation, the first element of the stack should move to the back k times and the order of the other elements should shift accordingly. You are not allowed to use the pop, push, or top methods. You may assume stack has at least 2 items. The solution should rotate the array in $O(N)$ time, where N is the number of elements in the queue. Assume that $k < N$.

Contents of the stack before (top is A):
A B C D E
Contents of the stack after ($k = 2$):
C D E A B